

# HOTFIRE

**H**ormonal **O**ptimization of Lung  
**T**ransplant Recipients **F**or  
**I**mprovements in **R**ehab and **E**xercise  
Capacity

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PRINCIPAL INVESTIGATOR

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New lungs don't bring back lost muscle. After transplant, recovery is limited by skeletal muscle, and the androgen axis that helps maintain it often stays suppressed. HOTFIRE follows every adult we evaluate to learn how often and why that happens. Within that cohort, a double-blind pilot of 20 listed men tests whether testosterone can be delivered safely from listing through 12 weeks after transplant, and estimates its effect on strength and walking.

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DESIGN

**Cohort + embedded RCT pilot**

RANDOMIZED

**20 men · 1:1**

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INTERVENTION

**Testosterone cypionate**

BLINDED TREATMENT

**Listing to 12 wk post-tx**

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SETTING

**Academic institution with  
~85 transplants / yr**

FOLLOW-UP

**Up to 24 months**

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1 · BACKGROUND

**Transplant replaces the lungs. It doesn't reset the axis.**

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## HR 7.5

FRAILITY, 1-YR DEATH

Muscle limits recovery after transplant, and pre-transplant frailty predicts death. In 386 recipients, frailty by SPPB carried an adjusted 1-year mortality HR of 7.5 (95% CI 1.6–36.0).

Singer 2018 · peripheral muscle oxidative capacity stays low after transplant (Wang 1999)

## 89%

HYPOGONADOTROPIC

Hypogonadism in lung disease is mostly central. 41.8% of men with COPD were hypogonadal vs 10.0% of controls, and nearly 9 in 10 had low or inappropriately normal gonadotropins.

Castro 2024 · testosterone tracks  $\text{PaO}_2$  (Semple 1984) and falls with steroid dose,  $r = -0.78$  (MacAdams 1986)

## IPF

NOT JUST COPD

The premise holds in fibrosis, which matters at an IPF-heavy center. Men with IPF have lower testosterone and DHT than age-matched controls, and testosterone tracks leukocyte telomere length. Genetically higher testosterone is associated with lower IPF risk (OR 0.41, 95% CI 0.19–0.91), with no reverse effect.

Fang 2020 · Xu 2024 · about a third of chronic ILD patients are sarcopenic (Sheehy 2024). All observational or genetic — no interventional trial exists.

## 41%

STILL LOW AFTER

In 51 men tested before and after transplant, mean testosterone went from 330 to 355 ng/dL ( $p = 0.48$ ). The share below 300 fell only from 51% to 41%.

Thompson 2025 · indication-driven testing, single unstandardized draws, no LH/FSH

## +3.5%

LEAN MASS ON  
PREDNISONE

Steroid-treated patients respond to androgens. Men on about 12.6 mg/day prednisone gained 3.5% lean mass on testosterone vs  $-0.9\%$  on placebo, with better strength. In COPD rehab, nandrolone restored the training response in steroid users: +20 W vs +5 W peak workload.

Crawford 2003 · Creutzberg 2003

### BEFORE TRANSPLANT

- Hypoxemia
- Systemic inflammation
- Intermittent steroids
- Cachexia, deconditioning



### AFTER TRANSPLANT

- **Daily prednisone**
- **Calcineurin inhibitors**
- **Critical illness, opioids**
- **Kidney injury**

Transplant removes hypoxemia but adds a new set of axis suppressors, so hypogonadism predictably persists. Every recipient takes glucocorticoids, which

makes the steroid-treated trials, not the COPD trials, the closest evidence for this study. An IPF-heavy cohort sharpens the contrast: unlike COPD patients, who often arrive on maintenance steroids, IPF patients generally are not on chronic glucocorticoids before transplant but all are after it.

3 · STUDY DESIGN

## Two linked components, analyzed separately

COMPONENT A · OBSERVATIONAL COHORT

Everyone we evaluate, men and women

Defines how often hypogonadism occurs, whether it is central or primary, and how it changes through 24 months. Tests whether post-transplant testosterone tracks cumulative prednisone and recovers as steroids taper.

COMPONENT B · RANDOMIZED PILOT

Listed men with confirmed, deferrable hypogonadism

1:1 to testosterone cypionate or matched placebo from randomization through 12 weeks after transplant, with usual care and rehab. Stratified by maintenance oral steroids at listing. Pharmacy holds the allocation; an unblinded titrator gives placebo patients sham dose changes. Everyone is unblinded once T3 is complete and offered open-label treatment.

PATHWAY TO RANDOMIZATION

- **Transplant evaluation**  
Adults 18 and older consent to the cohort.
- **Baseline hormone panel**  
Total and free T, SHBG, LH, FSH, estradiol.
- **Men: total T below 300 ng/dL twice**  
Fasting morning samples on separate days, with symptoms, outside acute illness. Done during evaluation, not after listing.
- **Listed · endocrinology confirms**  
Diagnosis confirmed and treatment can safely be deferred. Confirmation is finished before listing so randomization isn't the bottleneck.

Otherwise → clinical care, stays in Component A

REGIMEN (ADAPTED FROM FORTIFY)

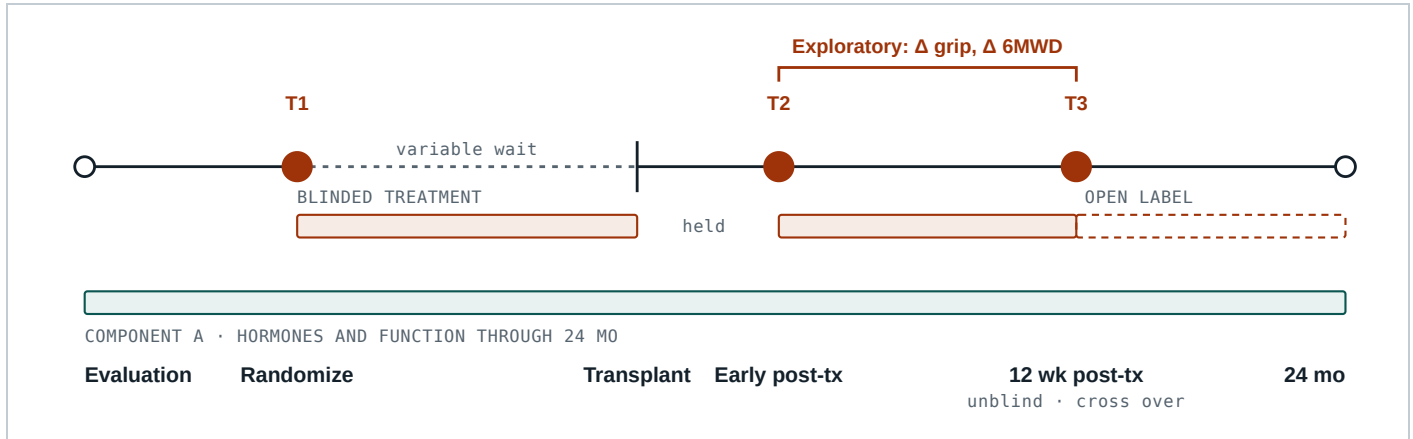
|          |                                                                      |
|----------|----------------------------------------------------------------------|
| Duration | Randomization through 12 wk post-transplant                          |
| Start    | 100 mg IM weekly                                                     |
| Week 6   | Trough T: below 400 → 150 mg; above 700 → 75 mg                      |
| Route    | SC at same dose if anticoagulated or platelets below 50,000/ $\mu$ L |
| Hold     | Hct above 54%, new VTE, AF needing treatment, AKI stage 2+           |
| At T3    | Unblind; continue or cross over to open label                        |

Usual prehab and post-transplant rehab in both arms, logged by session, exercise type and attendance.

**○ Randomize 1:1, double-blind**  
Stratified by maintenance oral steroids.

5–6 · TREATMENT AND ASSESSMENTS

## One participant, start to finish



Blinded treatment runs from randomization at listing through 12 weeks after transplant. Drug is held from the call to transplant and restarts once the patient is extubated 48 hours, off vasopressors and ECMO, not bleeding, and cleared by the transplant team; that visit opens T2. Once T3 assessments are complete, everyone is unblinded: patients on active drug may continue it with endocrinology, and those on placebo may cross over. With the median wait now under a month, most exposure falls after transplant — roughly 2–4 weeks listed, a 2–6 week hold, then about 10 weeks to T3, for 12–14 weeks in total.

### Why hold after surgery

Anabolic treatment in acute critical illness has backfired before: growth hormone roughly doubled ICU mortality (Takala 1999). Testosterone also raises AF, AKI and PE risk, all common right after transplant.

### Why T2 → T3

New lungs push 6MWD up regardless of drug. Starting the clock once patients are stable isolates muscle recovery. T1 → T3 stays secondary.

## Feasibility first, with a muscle signal

### COMPONENT A

Define prevalence, type and course of hypogonadism from evaluation to 24 months.

- A1: most hypogonadism is hypogonadotropic at evaluation
- A2: post-transplant hypogonadism tracks with cumulative prednisone

### COMPONENT B • PRIMARY

Show that a blinded trial can recruit, retain, dose and stay blinded from listing through 12 weeks after transplant, safely.

### EXPLORATORY EFFICACY OUTCOMES

Between-group difference in change from T2 to T3 in **grip strength** (kg) and **6MWD** (m). At 10 men per arm these estimate direction and variance; they cannot test efficacy. Both are tracked because anabolic trials in lung disease move strength more reliably than walk distance (Liu 2022; Pan 2014).

*A rise in testosterone without better strength or walking will not count as success.*

## Who enters the randomized pilot

### Inclusion • all required

Man 18 or older, formally listed at the center

Symptoms plus total T below 300 ng/dL on two fasting morning samples

Endocrinology confirms diagnosis and that treatment can be deferred

No testosterone or anabolic agent in the prior 3 months

Can do grip and 6MWD at listing; can consent

### Exclusion

Prostate, breast or other hormone-sensitive cancer

PSA above 4 ng/mL (3 if higher risk) or nodule without urology clearance

Hematocrit above 50%

MI, stroke or unstable angina within 6 months; decompensated heart failure

Prior unprovoked VTE or known thrombophilia

Transplant type is not an eligibility criterion. Single and bilateral recipients are both included, transplant type is recorded for everyone, and outcomes are reported for each group separately and combined.

7 · OUTCOMES

What gets measured

|                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Feasibility</b></p> <ul style="list-style-type: none"><li>• Eligible men randomized</li><li>• Retention to T3 or study exit</li><li>• Doses given vs scheduled</li><li>• Days to transplant, and total blinded days</li><li>• Restart by postoperative day 30</li><li>• T2 and T3 data completeness</li><li>• Blinding guess at T3, before unblinding</li><li>• Open-label uptake and crossover</li></ul> | <p><b>Exploratory efficacy</b></p> <p>Between-group difference in change from T2 to T3. At 10 men per arm these estimate direction and variance.</p> <ul style="list-style-type: none"><li>• <math>\Delta</math> grip strength, T2 <math>\rightarrow</math> T3</li><li>• <math>\Delta</math> 6MWD, T2 <math>\rightarrow</math> T3</li></ul> |
| <p><b>Prespecified safety</b></p> <ul style="list-style-type: none"><li>• Atrial fibrillation or flutter</li><li>• AKI, KDIGO stage 2+</li><li>• Venous thromboembolism</li><li>• Hct above 54%, PSA rise above 1.4 ng/mL</li><li>• Rejection, infection, tacrolimus out of range</li><li>• MI, stroke, death</li></ul>                                                                                         |                                                                                                                                                                                                                                                                                                                                             |

The first three safety events are the ones testosterone increased in TRAVERSE (AF 3.5% vs 2.4%, AKI 2.3% vs 1.5%, PE 0.9% vs 0.5%), and all three are common after lung transplant (Lincoff 2023; VTE 29%, Kahan 2007).

SECONDARY OUTCOMES

| OUTCOME                      | HOW IT IS MEASURED | WHEN |
|------------------------------|--------------------|------|
| MUSCLE AND PHYSICAL FUNCTION |                    |      |

| OUTCOME                               | HOW IT IS MEASURED                                                      | WHEN                        |
|---------------------------------------|-------------------------------------------------------------------------|-----------------------------|
| <b>Grip strength</b>                  | Hand dynamometer, dominant hand, best of three                          | T1 · T2 · T3 · 6, 12, 24 mo |
| <b>Six-minute walk distance</b>       | ERS/ATS field walking test standard                                     | T1 · T2 · T3 · 6, 12, 24 mo |
| <b>Quadriceps strength</b>            | Hand-held dynamometer, mean of three efforts                            | T1 · T2 · T3 · 6, 12, 24 mo |
| <b>Physical performance</b>           | Short Physical Performance Battery (SPPB)                               | T1 · T2 · T3 · 6, 12, 24 mo |
| <b>Time to independent ambulation</b> | Days to walking without another person's help; assistive device allowed | Daily while inpatient       |
| <b>PATIENT-REPORTED</b>               |                                                                         |                             |
| <b>Fatigue</b>                        | FACIT-Fatigue                                                           | T1 · T2 · T3 · 6, 12, 24 mo |
| <b>Quality of life</b>                | LT-QOL, built and validated in lung transplant recipients               | T1 · T2 · T3 · 6, 12, 24 mo |
| <b>BODY COMPOSITION</b>               |                                                                         |                             |
| <b>Lean body mass</b>                 | DEXA, already obtained in all patients during transplant evaluation     | T1 · T3                     |
| <b>CLINICAL RECOVERY</b>              |                                                                         |                             |
| <b>ICU and hospital stay</b>          | Days from transplant to ICU and to ward discharge                       | Index admission             |
| <b>Mechanical ventilation</b>         | Hours of invasive ventilation after transplant                          | Index admission             |
| <b>Discharge destination</b>          | Home vs rehabilitation or skilled nursing facility                      | Discharge                   |
| <b>Readmission</b>                    | Any-cause admission after transplant discharge                          | 90 days                     |
| <b>Rehabilitation exposure</b>        | Sessions attended, aerobic and resistance exposure, interruptions       | Throughout                  |
| <b>HORMONAL</b>                       |                                                                         |                             |
| <b>Testosterone and estradiol</b>     | Fasting, drawn before 10 am, at trough before the weekly dose           | Per schedule of assessments |
| <b>Target attainment</b>              | Proportion with trough testosterone 400–700 ng/dL                       | Week 6 · after each change  |

## 8 · STATISTICAL ANALYSIS

# Sized to test feasibility, not efficacy

**Precision.** With 10 men per arm, 80% power reaches only about 119 m in 6MWD or 6.6 kg in grip (assumed SDs of change 90 m and 5 kg) — far above anything the literature reports. Against plausible effects, power is roughly 11% for 30 m and 14% for 2 kg, and the 95% CI on the difference spans about ±85 m and ±4.7 kg. The pilot's job is to replace those assumed SDs with real ones. A definitive trial powered for 30 m needs about 140 per arm.

**Accrual.** 85 transplants a year × 57% men (this center) × 45% hypogonadal gives a pool of about 22 a year. Twenty randomized takes 1.8 years at 50% consent, 2.6 at 35%, 3.7 at 25%. With the median wait under a month, each patient reaches T3 about 4 months after randomization, so the trial closes ~4 months after the last enrollment.

**Functional analysis set.** Only men transplanted with both T2 and T3 contribute. Expect roughly 14 to 16 men, 7 to 8 per arm. All 20 contribute to feasibility, which is the point.

**Analysis.** ANCOVA of T3 adjusted for T2 and stratum, among transplanted men with data. Transplant and survival happen after randomization, so the report also gives full disposition by arm, a mixed model of all T1–T3 data, and multiple imputation.

**Component A.** Exact-CI prevalence; mixed model of testosterone with time-varying prednisone, tacrolimus, opioid, eGFR and BMI.

**Transplant type.** Single and bilateral recipients are both eligible. Functional, clinical and safety outcomes are reported separately for each and combined.

| PROGRESSION CRITERION                    | GO        | MODIFY                     | STOP              |
|------------------------------------------|-----------|----------------------------|-------------------|
| Eligible men randomized                  | ≥ 40%     | 25–39%                     | < 25%             |
| Randomized men retained to T3 or exit    | ≥ 85%     | 70–84%                     | < 70%             |
| Transplanted men with complete T2 and T3 | ≥ 80%     | 65–79%                     | < 65%             |
| Scheduled doses given                    | ≥ 80%     | 65–79%                     | < 65%             |
| Restarted by postoperative day 30        | ≥ 75%     | 50–74%                     | < 50%             |
| Safety board review                      | No signal | Fixable by protocol change | Unacceptable harm |

## Clinical care comes first

### Safety monitoring

An independent board (transplant pulmonologist, endocrinologist, statistician) reviews unblinded AF,

### Ethics

We randomize only when endocrinology judges treatment can wait. Anyone who needs testosterone gets it and stays in the cohort. The blinded period ends 12 weeks after transplant, when every

AKI and VTE rates every 6 months and after every 5 transplanted participants. Functional tests are run by assessors blinded to allocation and outside the patient's clinical care.

participant is unblinded and offered open-label testosterone whatever their assignment. Consent covers the TRAVERSE signals, sham dose changes, holds around surgery, fertility suppression, and the chance of never being transplanted.

FOR REVIEWERS

Defaults to confirm

| ITEM                         | DEFAULT IN THIS DRAFT                                                                                                                                                            | OWNER                     |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Accrual                      | Pool of ~22 eligible men a year (85 transplants, 57% men, 45% hypogonadal); 20 randomized in 1.8–3.7 years. Confirm the hypogonadism and consent assumptions against center data | PI                        |
| Treatment duration           | Blinded from randomization to T3, 12 weeks post-transplant, then unblinding and an open-label offer                                                                              | PI<br>Endocrinology       |
| Screening-to-listing timing  | Screening and endocrine confirmation finished during evaluation, so randomization isn't the rate-limiting step before a short wait ends in transplant                            | PI<br>Endocrinology       |
| Titration and Hct thresholds | Trough 400–700 ng/dL; exclude Hct above 50%, hold above 54%                                                                                                                      | Endocrinology<br>Pharmacy |
| Restart after transplant     | Extubated 48 h, off pressors and ECMO, no bleeding                                                                                                                               | Transplant                |
| Stratification               | Maintenance oral steroids at listing                                                                                                                                             | Biostatistics             |
| DEXA at T3                   | Evaluation scan is routine; confirm whether the week-12 scan is clinical or research-funded                                                                                      | Investigators             |
| Safety board                 | Transplant pulmonologist, endocrinologist, statistician                                                                                                                          | PI<br>IRB                 |

10 · SIGNIFICANCE

The first map of the axis across transplant

If post-transplant hypogonadism is a predictable effect of steroid-based immunosuppression, it becomes a screenable, treatable cause of poor recovery. A ClinicalTrials.gov search found no registered randomized trial of testosterone in

lung transplant; the nearest is a phase 1 kidney-candidate study with historical controls (NCT07287800).

#### KEY REFERENCES

1. Thompson A, et al. Post-transplant testosterone in men undergoing lung transplantation (TriNetX). *Int J Impot Res* 2025
2. Castro HM, et al. Prevalence of hypogonadism in men with and without COPD. *Endocrinol Diabetes Nutr* 2024
3. Fang C, et al. Sex hormones and leucocyte telomere length in men with IPF. *Respirology* 2020
4. Xu Q, et al. Testosterone, estradiol, estrogen sulfotransferase and IPF: bidirectional Mendelian randomization. *BMC Pulm Med* 2024
5. Sheehy R, et al. Sarcopenia in interstitial lung disease. *Eur Respir Rev* 2024
6. MacAdams MR, et al. Reduction of serum testosterone during chronic glucocorticoid therapy. *Ann Intern Med* 1986
7. Crawford BAL, et al. Androgen effects on muscle and bone in men on long-term glucocorticoids. *JCEM* 2003
8. Creutzberg EC, et al. Anabolic steroids in COPD rehabilitation. *Chest* 2003
9. Casaburi R, et al. Testosterone and resistance training in men with COPD. *AJRCCM* 2004
10. Liu Y, et al. Anabolic-androgenic steroids for COPD: meta-analysis. *Front Med* 2022
11. Singer JP, et al. Frailty phenotypes and mortality after lung transplantation. *AJT* 2018
12. Wang P, et al. Pulmonary rehabilitation after lung transplantation: meta-analysis. *J Thorac Dis* 2024
13. Lincoff AM, et al. Cardiovascular safety of testosterone replacement (TRAVERSE). *NEJM* 2023
14. Bhasin S, et al. Testosterone therapy in men with hypogonadism: Endocrine Society guideline. *JCEM* 2018
15. Eldridge SM, et al. CONSORT extension to randomised pilot and feasibility trials. *BMJ* 2016

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HOTFIRE · Draft protocol · Diego Marin, DO, FCCP

Not a treatment order or approved clinical protocol